Lec 4B: Virtualization & Cloud computing

Graduate Systems (CSE638)

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Content taken from:
- Design and Engineering of Computer Systems, Mythili Vutukuru (IITB)
https://www.cse.iitb.ac.in/~mythili/decs/
- CS 695: Virtualization and Cloud Computing, Mythili Vutukuru (IITB)
https://www.cse.iitb.ac.in/~mythili/virtcc/



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Virtualization and cloud computing

- What is the cloud?
  - Commodity servers with lots of compute and storage, connected with high speed networking, located in data centers
- What is virtualization?
  - Multiple virtual machines (VMs) can run inside a physical machine (PM)
  - VM gives user an illusion of running on a physical machine
  - Containers are like lightweight VMs
- Virtualization is a building block for cloud computing
  - Virtualization enables multiple clients share the cloud's compute resources
  - Multiple users on VMs/containers can share same cloud server
- In addition to compute, clouds also manage large amounts of data
  - Cloud storage/big data systems for efficient storage and retrieval of data

Why cloud computing?

- Public cloud providers (Amazon AWS, Microsoft Azure, Google Cloud etc) setup and maintain data centers with high-end servers
  - Powerful CPUs, lots of memory, disk storage etc., available to users
  - Organizations can also run a private cloud only for their users
- Why run applications on cloud and not on “bare metal” servers?
  - Multiplexing gains: multiple VMs can share the system resources
  - Lower overhead of maintenance: hardware/software maintained by providers
  - Flexibility: VMs can move to another machine if one fails
  - Pay as per usage: no need to invest in servers if only lightly used
- Disadvantages of running applications on cloud
  - Performance: longer delay to access servers via internet
  - Higher cost if heavily used

How is the cloud used? (1)

- Many ways to expose the cloud to users/developers
- Infrastructure as a Service (IaaS): Cloud provider exposes cloud infrastructure (VMs, storage, etc.) directly to cloud users.
  - Developer installs application of choice on the VMs, storage
  - Provider handles VM lifecycle management, placement of VMs on servers, migration in case of server failures, elastic scaling of VMs under load etc.
  - Hassle-free management of compute and storage
  - Flexible usage, pay-as-you-go model
- Examples
  - **Amazon EC2**: exposes virtual servers (VMs) to users
  - **Amazon Simple Storage Service (S3):** exposes basic cloud storage
  - **Microsoft Azure, Google Cloud, Digital Ocean** etc. also have similar offerings

How is the cloud used? (2)

- Platform as a Service (Paas): Cloud provider sets up software on the VMs and exposes the infrastructure + software bundle to users
  - Provider exposes APIs using which developers build cloud apps
  - Provisioning VMs and storage for the app, maintaining application software (OS, app servers), scaling application tiers etc. handled by the cloud provider
  - Developer only focuses on application logic
  - Easier to build cloud apps/software but lesser flexibility
- Examples
  - **Amazon Elastic Beanstalk:** provides APIs to build cloud apps, automatically deploys them on EC2 instances
  - **Google Cloud App Engine, Heroku** are other examples

How is the cloud used? (3)

- Software as a Service (Saas): Cloud provider installs useful software on the cloud and let users access the software directly
  - Users have no visibility into the cloud infrastructure
  - Users access the app software over the internet (nothing installed locally)
  - Application servers not hosted directly by the enterprise but on the cloud
- Examples
  - **Google docs, Gmail, Office 365**
  - Most internet-scale software companies (**Zoom, Dropbox, Netflix, Prime Video, Slack, Spotify, Whatsapp**) host their software on the cloud and not on their premises, for scalability, fault tolerance, and general ease of management

How is the cloud used? (4)

- Function as a Service (FaaS): Provider exposes only ability to run functions on cloud data
  - Also called serverless computing
  - Developers write functions to execute on certain triggers, e.g., function to convert a video to lower resolution is triggered whenever a video is uploaded
  - Provider instantiates VMs/containers to run the function when required
  - Users only pay when function runs, not for the infrastructure, so cheaper when functions are run sporadically
- Example: **AWS Lambda, Google Cloud Functions** etc.

Compute and data storage options

- Cloud compute runs on VMs or containers

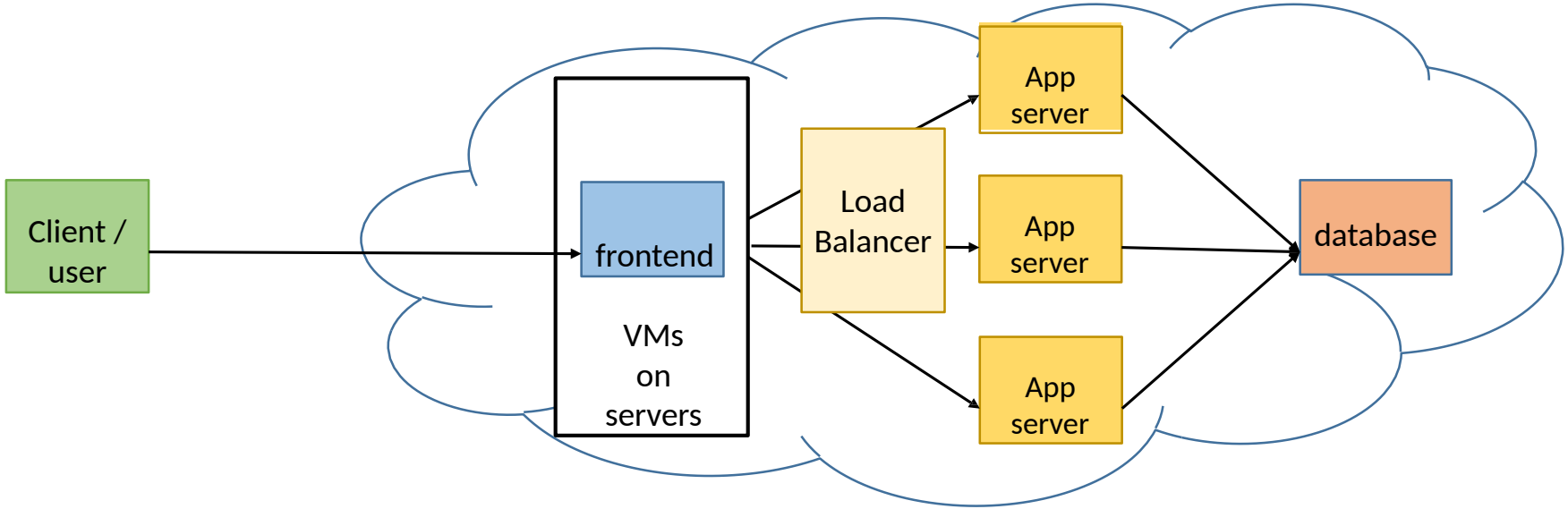
- Cloud storage options
  - Relational databases (RDBMS) with strong consistency (ACID properties). Traditional storage option, but harder to scale, lower performance.
  - No-SQL data stores map key to unstructured “blob” of data. Simpler get/put interface using single key, no transaction support. Some form of weaker consistency. Tradeoff stronger guarantees for higher performance.
  - Semi-structured data stores map key to value which has certain column families, but not fixed schema. Useful for certain apps.
  - Other storage frameworks customized for application needs

Cloud storage systems: examples from real-life

- Dynamo: high-performance No-SQL key-value store from Amazon
  - Trade off consistency guarantees for high availability and scalability
- Bigtable: semi-structured data store used at Google
  - Built over two other systems from Google: Chubby (a distributed locking/consensus system) and the Google File System (a distributed file system)
- Haystack: cloud storage system for efficiently storing and retrieving photos at Facebook
- Memcached: distributed cache used at Facebook to cache queries from database servers

Cloud applications

- Users access cloud applications over the internet
  - Running on **public or private clouds** (not running on “baremetal”)
- Cloud apps designed as multi-tier apps hosted on VMs/containers on servers
- Auto-scaling, fault tolerance, optimal placement on servers, network config
  - Functionality available from cloud management software (e.g., OpenStack)



Cloud management & Orchestration software

Facilities management of VMs and containers on the cloud

- Lifecycle management of VMs
- VM placement
- VM migration
  - Server maintenance
- Instantiation of additional replica of components (auto-scaling)
- Examples: Openstack, Kubernetes

Virtualization

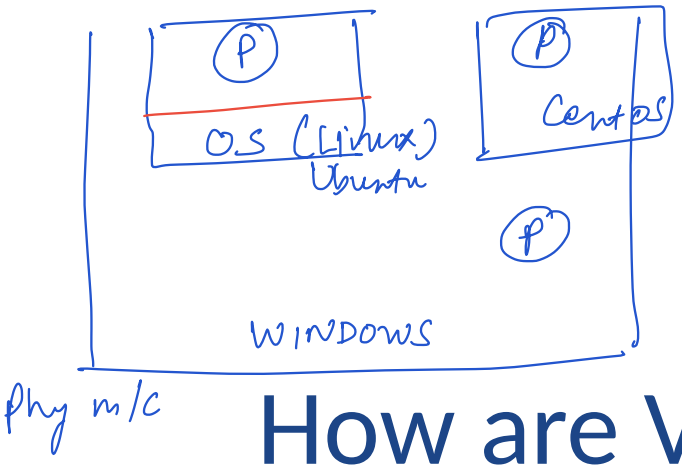
Story so far …

- User programs run over OS → OS virtualizes CPU, memory, network, I/O, …

Virtualize the entire system? One system = Multiple separate systems?

- Why?
  - Efficient hardware sharing
  - Better isolation (compared to processes)
- How?
  - Run Virtual machines (VMs) on the same Physical Machine (PM)
  - Containers (light-weight VMs)

System Virtualization → Primary requirement for Cloud computing



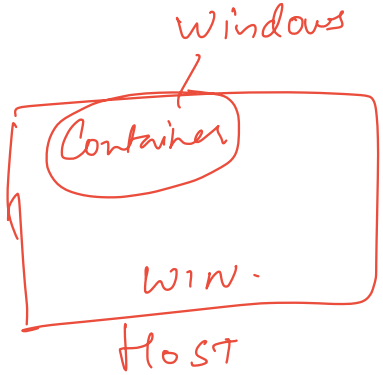
How are VMs implemented?

Virtual Machine Monitor (VMM)

CPU Mem I/O

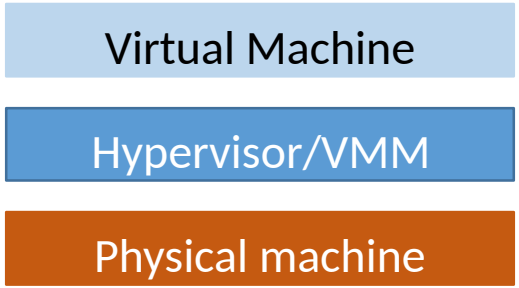
"Virtual Machines" &

"Containers"



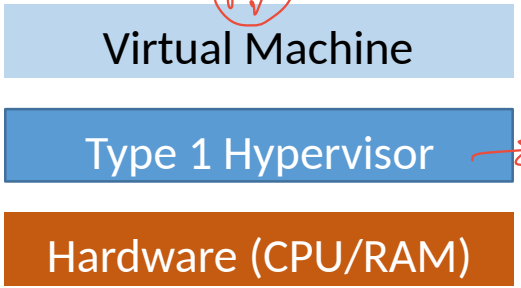
Virtualization terminology (1)

- We will study system virtualization, or how to run one full system (OS and applications) over another OS
- Hypervisor or virtual machine monitor (VMM): a piece of software that allows multiple VMs to run on a physical machine (PM)
  - We will study how VMMs are designed



Virtualization terminology (2)

- Guest OS runs inside the VM, and host OS runs on the PM
- Type 1 hypervisor: runs directly on hardware, no need for host OS



- Type 2 (hosted) hypervisor: runs as an application on top of host OS

